

Basics of Computer Graphics

Unit I Overview



Unit I: Core Concepts

In this unit, we will cover the foundational elements of computer graphics that are essential for understanding how digital images are created and manipulated.

1 Coordinate Systems

Understanding how points and objects are located in 2D and 3D space.

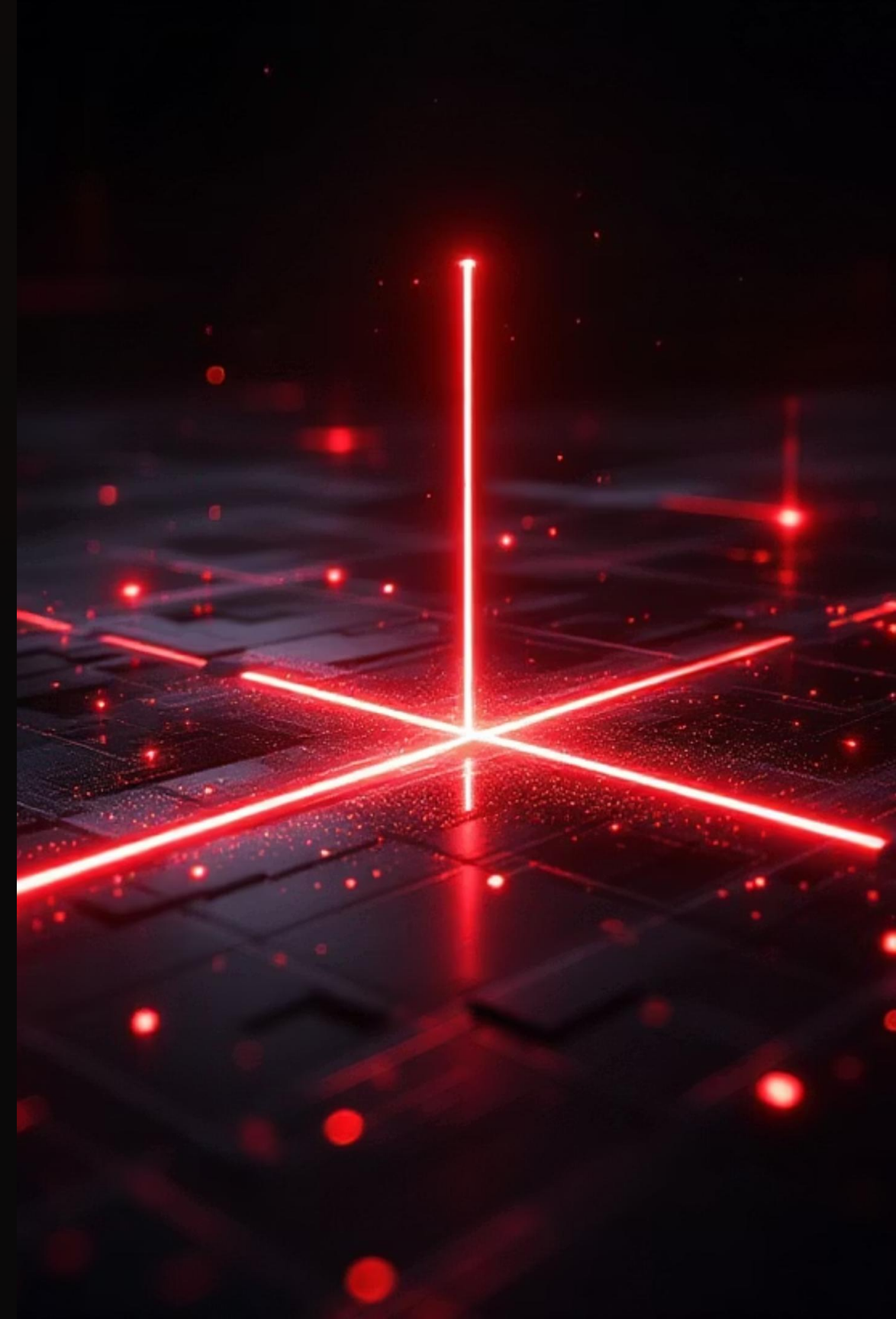
2 Graphics File Formats

Exploring the various ways images are stored and the implications of each format.

3 Graphics Functions & Standards

An overview of the programming interfaces and protocols that enable graphics rendering.

Coordinate Systems: Defining Space

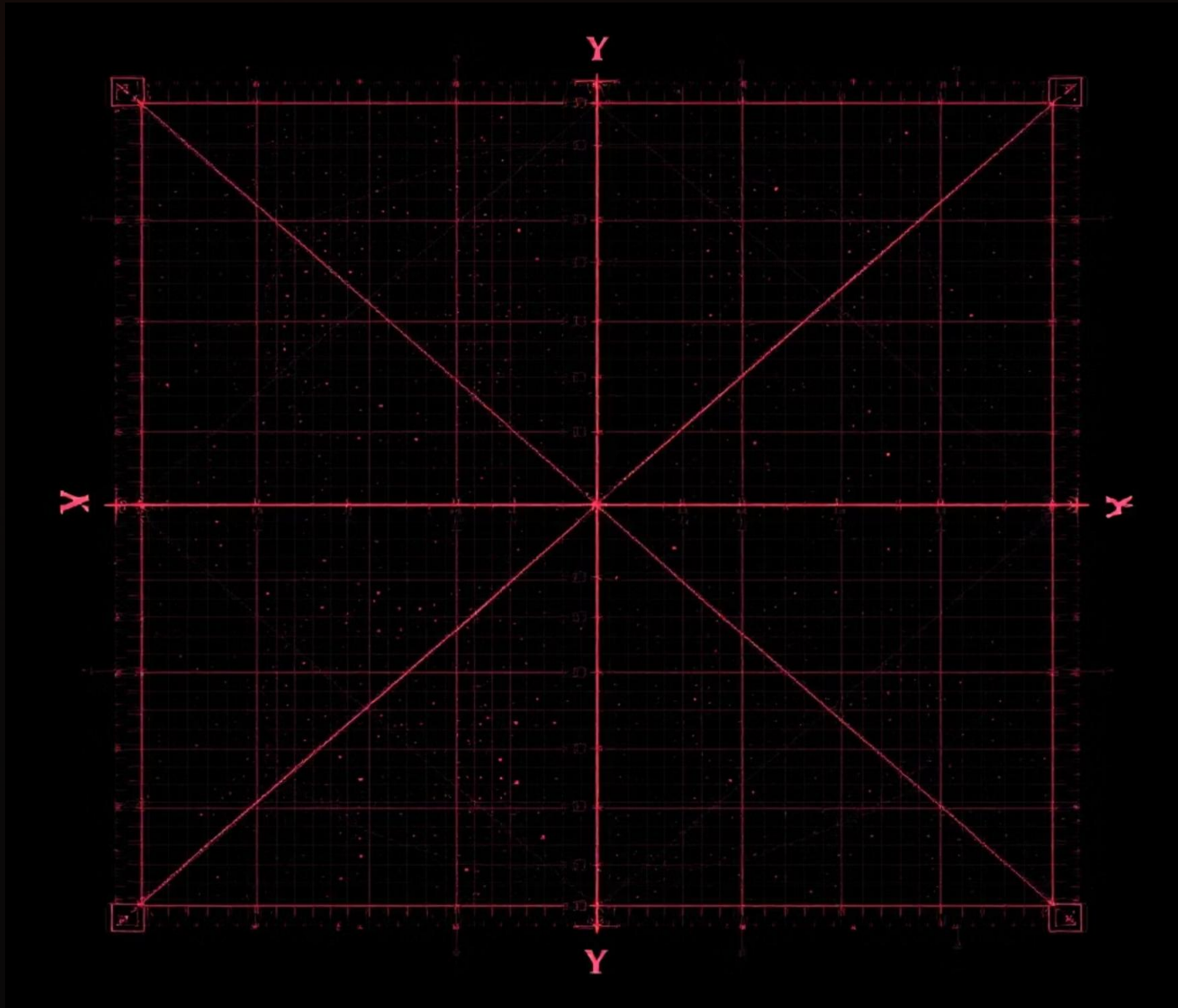


What is a Coordinate System?

A coordinate system is a framework used to define the position of points in space. It's fundamental to placing any object or pixel on a digital canvas.

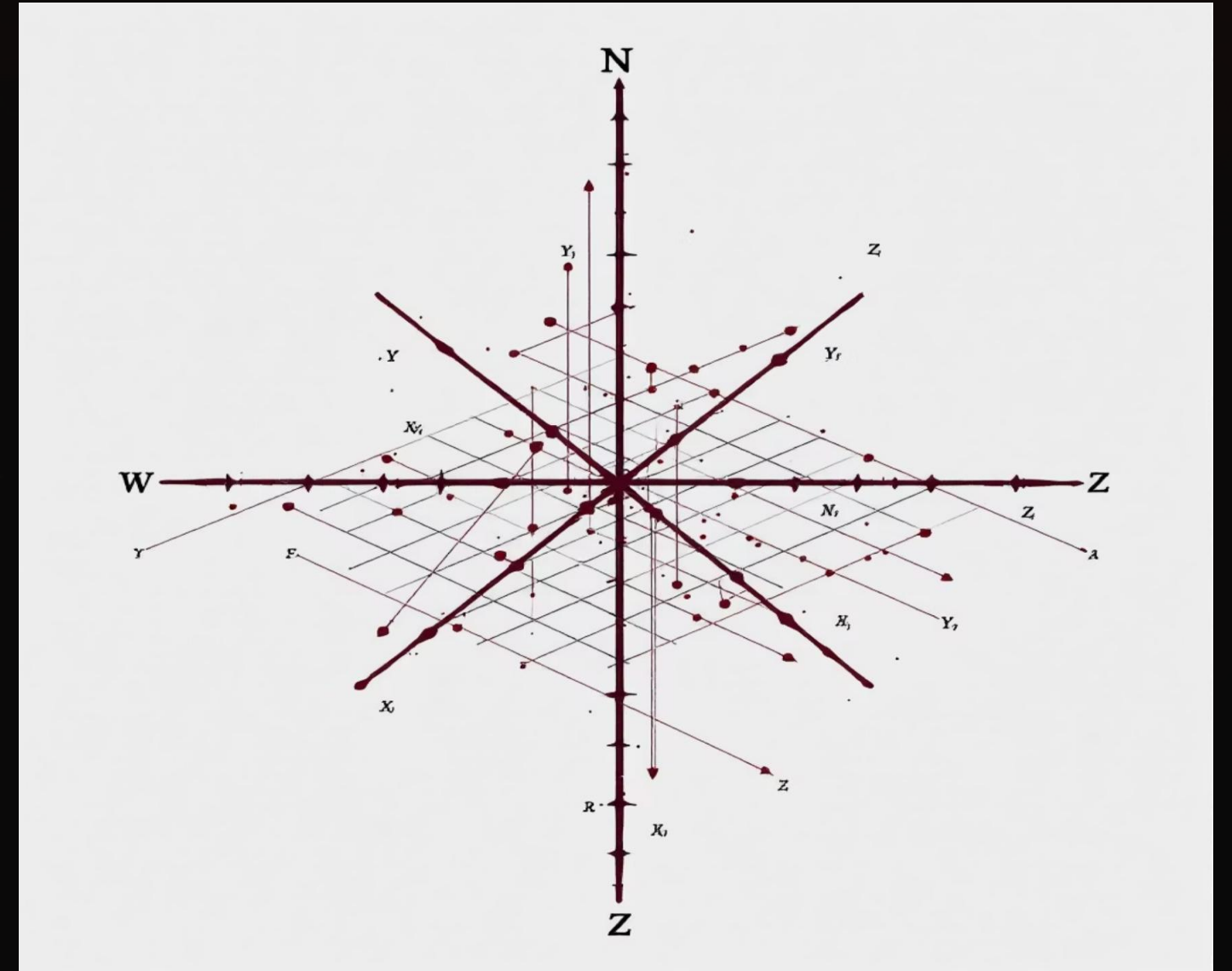
2D Coordinates

Uses X (horizontal) and Y (vertical) axes to locate points on a flat plane, common in basic image editing and UI design.



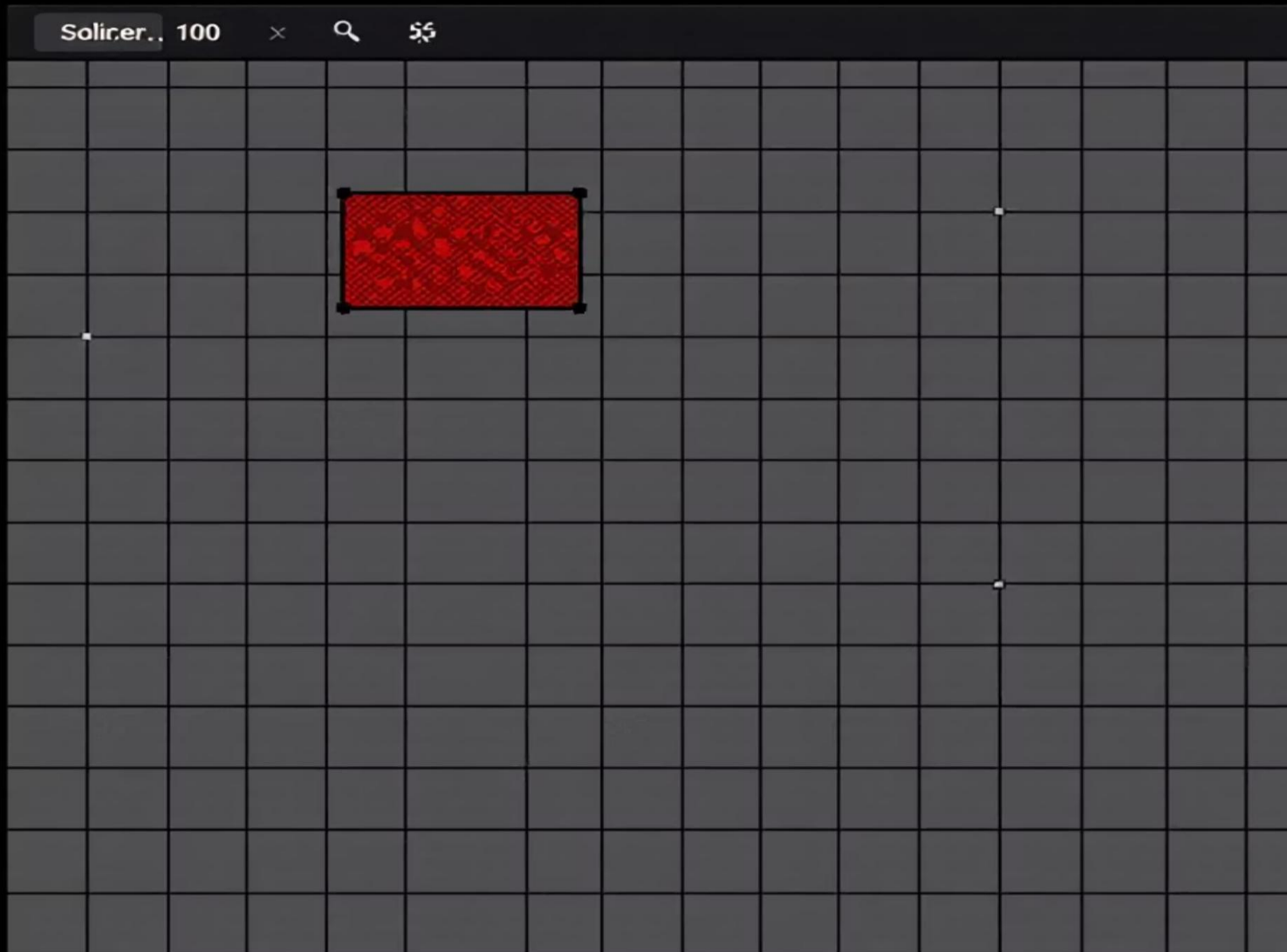
3D Coordinates

Adds a Z (depth) axis to 2D, allowing for the representation of objects in a volumetric space, crucial for games and 3D modeling.



2D Coordinate System in Practice

In many computer graphics applications, especially for 2D, the origin (0,0) is located at the top-left corner of the screen or canvas. The X-axis increases to the right, and the Y-axis increases downwards.

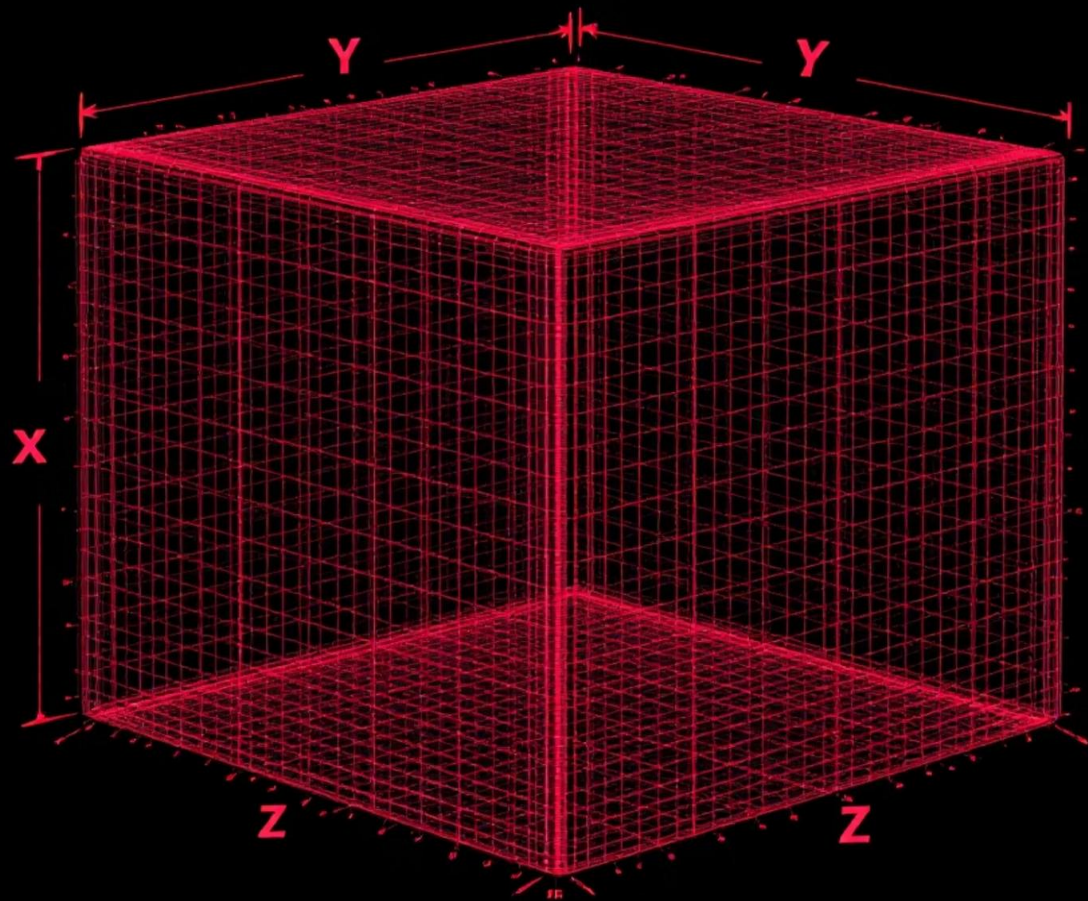


Drawing a Rectangle

- Origin: (0,0) at top-left
- X-axis: Increases rightward
- Y-axis: Increases downward
- Example: A rectangle starting at (50, 100) means it begins 50 pixels from the left edge and 100 pixels from the top edge.

The Z-Axis: Adding Depth to 3D

The Z-axis introduces the concept of depth, transforming a flat plane into a volumetric space. This allows objects to have a position not just left/right and up/down, but also closer or further away from the viewer.



Applications

- Used extensively in video games to create immersive environments.
- Essential for 3D modeling software, allowing designers to sculpt and manipulate objects in three dimensions.
- Critical for virtual reality (VR) and augmented reality (AR) experiences to place virtual objects within real-world space.



Graphics File Formats

The language of digital images.

What are Graphics File Formats?

Graphics file formats are standardized ways to organize and store digital images. Each format employs specific algorithms for compression and data representation, leading to unique advantages and disadvantages for different use cases.

Choosing the right format depends on the image's intended use, balancing quality, file size, and feature support.

Graphics File Format Comparison

BMP	Uncompressed, high fidelity, excellent for retaining original image data.	Extremely large file sizes, impractical for web or limited storage.
GIF	Supports animation, transparent backgrounds, small file sizes for simple graphics.	Limited to 256 colors, unsuitable for photographs or gradients.
JPEG	Excellent for photographs with millions of colors, highly compressed, small file sizes.	Lossy compression means quality degrades with each save, not ideal for sharp lines or text.
TIFF	High quality, supports various color depths, ideal for print, professional publishing, and scanning.	Very large file sizes, can be slow to load, less common for web use.
PCX	Simple and efficient for early DOS applications, supported various color depths.	Primarily outdated, limited modern software support, less efficient than newer formats.

Examples

Understanding where each format excels helps in making informed decisions for your graphics projects.

BMP: Simple Graphics



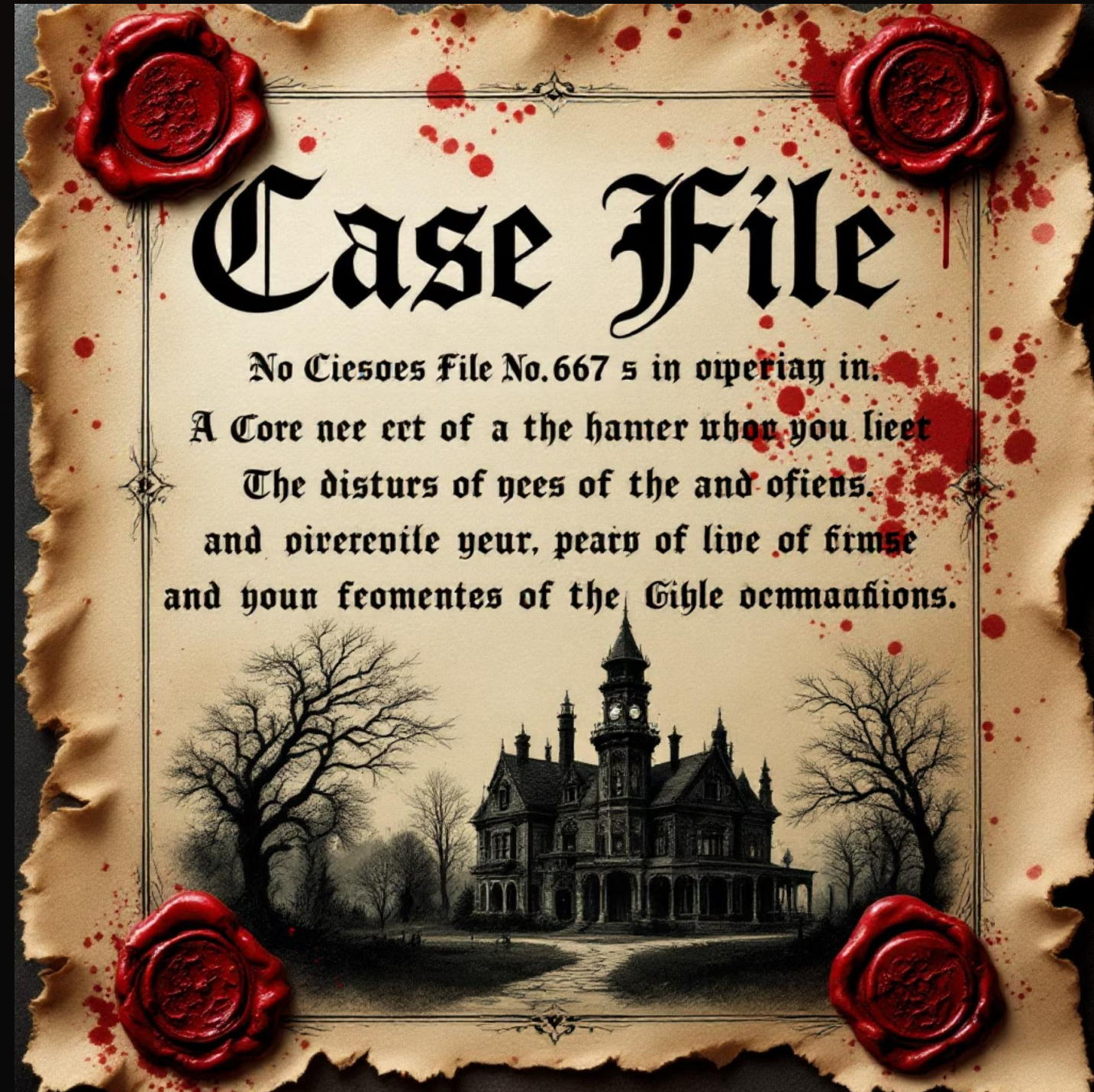
JPEG: Photography



GIF: Animated & Simple Icons



TIFF: Print & Scanning



Examples i

PCX: Classic Gaming



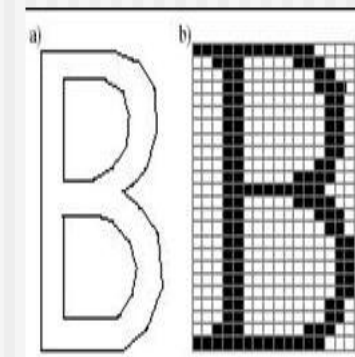
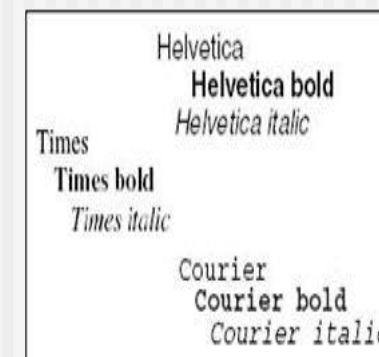
Text mode

- Text mode is a personal computer display setting that divides the display screen into **25 rows and 80 columns** in order to display text without images.
- In text mode, each box can contain one character. Text mode contrasts with graphics mode, which features an array of pixels instead of text boxes.
- Text mode is also known as **character mode** or **alphanumeric mode**.

Text

- Some graphics devices have two distinct display modes.
 - . Text mode
 - . Graphics mode
- A routine to draw a character string might look like
`drawString(x, y, string)`

There are many text attributes, the most important of which are the text's font, color, size, spacing, and orientation.



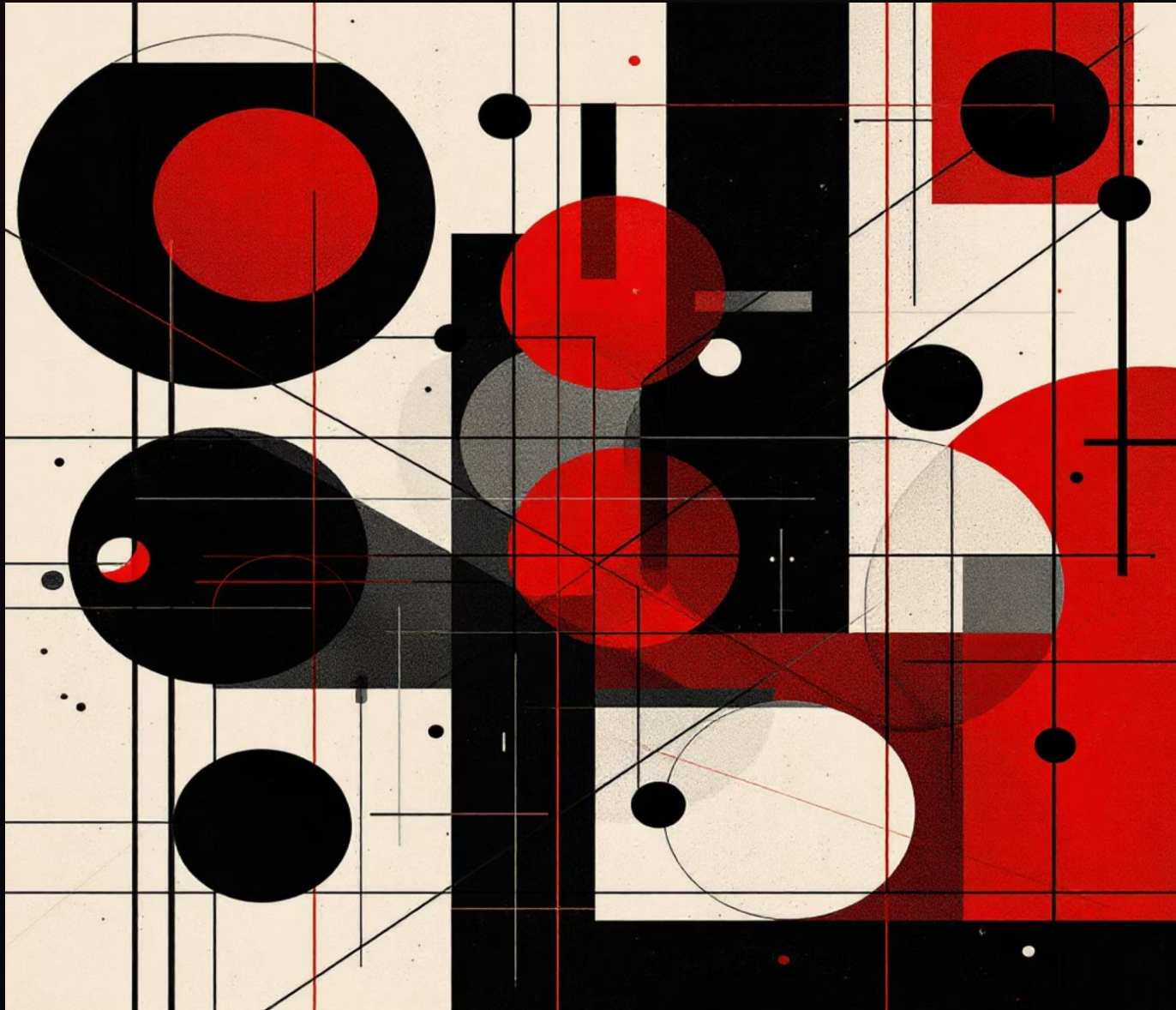
Graphics mode

- **Graphics mode** is a computer display mode that generates image using pixels.
- Today, most users operate their computer in a graphics mode opposed to a text mode or command line environment.
- Here, Screen is divided into small dots called pixels.
- e.g. VGA screen resolution is 640×480 pixels.
SVGA screen resolution is 800×600 pixels.



Shapes in Graphics

At the fundamental level, all complex visual elements in computer graphics are constructed from basic geometric shapes. These simple building blocks are essential for everything from primitive drawing to sophisticated user interface design.



Core Geometric Primitives

- **Lines:** The most basic element, connecting two points. Used for outlines, edges, and defining paths.
- **Rectangles:** Defined by two points (top-left and bottom-right) or a point, width, and height. Common for buttons, windows, and bounding boxes in UI design.
- **Circles/Ellipses:** Defined by a center point and radius (or two radii for ellipses). Used for circular elements, avatars, and radial menus.
- **Polygons:** Generalization of rectangles, defined by a series of connected points. Used for irregular shapes and complex object definitions.

Colors in Graphics

Understanding how colors are represented and manipulated is fundamental in computer graphics. The most common model for digital displays is the RGB color model.

The **RGB model** is an additive color model in which Red, Green, and Blue light are combined in various ways to reproduce a broad array of colors. It's the basis for how screens display images.

How RGB Colors Mix:

- **Red + Green = Yellow**
- **Red + Blue = Magenta**
- **Green + Blue = Cyan**
- **Red + Green + Blue = White** (when all are at full intensity)
- **No Red, Green, or Blue = Black** (absence of light)

Each color component typically ranges from 0 to 255, allowing for over 16 million possible colors ($256 \times 256 \times 256$).



Graphics Standards

Graphics standards define the interfaces and protocols that allow software applications to communicate with graphics hardware. They ensure consistency and efficiency in rendering visual content across different devices and platforms.

OpenGL (Open Graphics Library)

- A cross-platform, multi-language API for rendering 2D and 3D vector graphics.
- Widely used in CAD software, virtual reality, scientific visualization, and video games.
- Known for its flexibility and lower-level control, giving developers more power over the rendering pipeline.

DirectX

- A collection of APIs for handling tasks related to multimedia, especially game programming and video.
- Primarily used on Microsoft Windows platforms and Xbox consoles.
- Provides high-performance 2D and 3D graphics rendering, input, and sound.

Key Graphics API Logos

